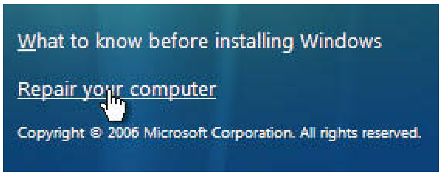
2. Most Linux distributions will also give you a recovery option right at the bootloader (Grub) stage. Try that and then do the same as above.
3. For Windows, if Safe Mode doesn't work, try using the "Last Known Good Configuration" option.
4. Since Windows 7, the System Restore tool has become a decently mature tool. You can access System Restore via the recovery disk as explained above, where instead of "Startup Repair" you can choose "System Restore". If you don't have the disc and can access the command prompt from the F8 menu, then log in as administrator and type `%systemroot%\system32\restore\rstrui.exe` to start system restore.

If at this point your system still doesn't start, you might need to reinstall the OS in the case of Windows (Don't worry, your data will still be safe).



Repair

Boots into the OS, some hardware not working

In case you have a piece of hardware which was previously working but has now stopped working, try the following:

1. Check that piece of hardware on another machine to see if it's a hardware issue or a software issue. If you lack access to another machine, booting into an Ubuntu Live CD and seeing if the hardware works there can provide a good workaround.
2. If it works, then it's most probably a driver update issue. Go to `Control Panel>Device Manager`, then select the malfunctioning device and click on Update Driver. If this doesn't fix the issue, go to the respective manufacturer's web site, download the latest device driver from there and install it.

Boots into OS, non-optimum performance

1. If you can't find a partition, type Disk Management in the Start menu and explore what the issue is. If you're unable to get to this stage, then you can pop in the Ubuntu Live drive and Open the "Gparted Tool" to investigate.
2. Blue Screen of Death: If you consistently find yourself running into the Blue Screen of Death here's what to do.
   a. On Windows 7 or later a consistent BSOD typically indicates a hardware or driver issue. As mentioned above, try updating your drivers.
   b. Run the MemTest86+ tool and see if your RAM is performing upto the mark. RAM is often the cause of consistent BSODs.
   c. One annoying thing is that often the system restarts without giving you time to note down the error, which you can later Google to get towards a reasonable solution. For this Right Click on your "My Computer" Icon and select "Properties", Click on "Advanced System Settings" from the right hand side panel. Click on "Settings" at bottom in the "Startup and Recovery" area and simply uncheck the "Automatically Restart" box under System Failure. This will give you all the time you need to note down the error.
3. Anti-Virus and spyware:
   a. A free antivirus software is a good first line of defence. Install one of them and keep the definitions regularly updated.
   b. Use SuperAntiSpyware & MalwareBytes on a regular basis with updated definitions to keep your system running smooth and trojan-free.
4. Slow startup
   a. In the Windows 7 menu, type "`msconfig`" by clicking on the Start Button, or in an older version, in the run command box. From there select "Startup" in the tab and uncheck all the software you don't want to start automatically.

TOOLS YOU'LL NEED

Here's a small list of tools that we've mentioned and used in this workshop. Most software will be found on the Digit DVDs, past and present.

Screwdriver Set: If you want to open your desktop cabinet or reseat your laptop RAM you'd need this

Ubuntu Live Drive: Even if you're a Windows user it will be an invaluable tool. It comes pre-bundled with a great partition manager known as "Gparted". Simply get one from http://dgit.in/GMDotz. It's recommended to use a USB stick

SuperAntiSpyware and MalWareBytes: Get these from http://www.superantispyware.com/ & http://www.malwarebytes.org/

Memtest86+ Get it here: http://www.memtest.org/#downiso

Windows 7/Vista/XP setup or recovery disk: If you're still using an OS released 12 years ago, consider an upgrade.

BlueScreenView: Available at http://dgit.in/GJydOc Will help you a great deal in debugging Blue Screen of Death errors.

Soluto: Obtained from https://www.soluto.com/ for automated system maintenance

5. Install Soluto for automated application updates and general maintenance.

Prevention and basic maintenance

Hopefully this guide would help you recover your system in the event of a crash with a little bit of smart Googling, a bit of patience and some common sense. We would like to wind up this guide by iterating the old adage which says, "prevention is better than cure". Here are a few simple tips for keeping your system well-oiled and preempting crashes.

- Defragment your system often.
- Keep your Registry clean by using tools like "CCleaner"
- Use the built-in "Reliability Monitor" tool in Windows 7 & 8.
- Keep backup images of your OS partition from a point where everything was working and updated.